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The Analysis of Rates and of Survivorship Using Log-Linear Models

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SUMMARY

Models are considered in which the underlying rate at which events occur has a log-linear relationship with covariates. It is shown that the estimation of parameters involves the solution of identical systems of equations for data from either a Poisson process, an exponential distribution, a survival model or a generalized log-linear model. This enables one to use algorithms for fitting log-linear models, such as iterative proportional fitting (IPF), for the analysis of rates or survivorship.

1. Introduction

Two types of data which arise from medical or epidemiological investigations are: data on rates and data on survivorship. The sources of data may be official vital statistics or disease registries which may involve a large number of individuals. One type of statistical summary is a set of rates [(number of events)/(total length of exposure)] computed for the population broken down by several factors (e.g. age, race and sex). A second type of data deals with the survival experience of the population, where the summary is based on the life table and individuals are again categorized by several variables.

Log-linear models have proved to be useful for analyzing categorical data. In this communication we show how log-linear models may be used in the context of both rates and survivorship.

2. Models for Rates

Bishop, Fienberg and Holland (1975, pp. 131–136) discuss the application of log-linear models to rates where the rate for the subgroup i $(i = 1, \ldots, I)$ of the population is $\lambda_i = n_i/T_i$. In this case, T_i is the total population size for subgroup i and n_i is the number in i who exhibited the event. However, in other contexts T_i may be a measure of total length of time that the population was under observation.

One model which has been described by Armitage (1966) assumes that n_i has a Poisson distribution with mean $m_i = \lambda_i T_i$. We shall assume that λ_i has a log-linear relationship with the vector of J variables, $\mathbf{X}_i$, thus

$$\lambda_i = \exp(\alpha + \mathbf{X}_i'\boldsymbol{\beta}). \tag{2.1}$$

This yields a multiplicative model for the rates, which has also been postulated by Kilpatrick (1962), Bjarnason *et al.* (1974), Breslow and Day (1975), Osborn (1975) and Gail (1978); Haberman (1978, pp. 60–62) discusses the application of log-linear models to this case.

Key words: Log-linear models; Iterative proportional fitting; Survival analysis; Vital rates.

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This model may be obtained in another way. Suppose one is concerned with times until failure, t_{ih}, where h indexes the individuals ($h = 1, \ldots, H_i$) and t_{ih} has an exponential distribution with parameter λ_i. A model where λ_i takes the log-linear form given in (2.1) has been described by Feigl and Zelen (1965) and by Glasser (1967). The data consist of observed follow-up times t_{ih}, which may be censored. Sufficient statistics for the parameters are $T_i = \sum_h t_{ih}$ and n_i, the number in the ith subgroup who were observed until the event occurred. We let $m_i = \lambda_i T_i$, which is not a mean as in the Poisson case, but is an 'expected frequency' produced by contingency table programs.

The following theorem relates the maximum likelihood estimates of $\boldsymbol{\beta}$ for the Poisson and exponential distributions to a multinomial distribution.

Theorem. Suppose that $\lambda_i = \exp(\alpha + \mathbf{X}_i'\boldsymbol{\beta})$, where $\mathbf{X}_i' = (X_{i1}, \ldots, X_{iJ})$ and $\boldsymbol{\beta}' = (\beta_1, \ldots, \beta_J)$. The maximum likelihood estimators for $\boldsymbol{\beta}$ are equivalent for (i) $n_i \sim \text{Poisson}(\lambda_i T_i)$ $(i = 1, \ldots, I)$; (ii) $t_{ih} \sim \text{exponential}(\lambda_i)$ $(i = 1, \ldots, I$ and $h = 1, \ldots, H_i)$; and (iii) $\mathbf{n}_{I\times 1} \sim \text{multinomial}(\mathbf{P}_{I\times 1}, n_+)$, where

$$P_i = \frac{T_i}{\sum_i T_i} \exp(\alpha + \mathbf{X}_i'\beta), \quad i = 1, \ldots, I,$$
$$n_+ = \sum_i n_i \quad \text{and} \quad \sum_i P_i = 1.$$

Proof. The kernel for the log likelihood for (ii) is

$$L = n_+\alpha + \sum_i n_i \mathbf{X}_i'\boldsymbol{\beta} - \sum_i \exp(\alpha + \mathbf{X}_i'\boldsymbol{\beta}) T_i, \tag{2.2}$$

where $T_i = \sum_h t_{ih}$. Taking derivatives of (2.2) with respect to α and $\boldsymbol{\beta}$ and setting them equal to zero, we find the following J equations which must be solved in order to find the maximum likelihood estimators (MLE) $\hat{\boldsymbol{\beta}}$:

$$\sum_i n_i X_{ij} - \frac{n_+ \sum_i X_{ij} T_i \exp(\mathbf{X}_i'\hat{\boldsymbol{\beta}})}{\sum_i T_i \exp(\mathbf{X}_i'\hat{\boldsymbol{\beta}})} = 0, \qquad j = 1, \ldots, J. \tag{2.3}$$

Similarly, the MLE for $\boldsymbol{\beta}$ from distributions (i) and (iii) can be shown to be solutions of (2.3) (see Bishop *et al.*, 1975, p. 447). Hence, all these maximum likelihood estimators of $\boldsymbol{\beta}$ are equal.

This theorem shows that MLE of $\boldsymbol{\beta}$ from the Poisson and exponential models can be found by using algorithms which are based on multinomial frequencies n_i with expectation $\lambda_i T_i$ for known T_i. Two algorithms which are commonly employed for such models are iterative proportional fitting (IPF) and Newton–Raphson. In both cases one can use algorithms based on the multinomial simply by specifying the model in the appropriate way.

Darroch and Ratcliff (1972) showed how IPF can be used to find the MLE for the general multinomial model described above. The theorem shows that IPF can also be employed for our models which incorporate time. The steps to be used are: (i) take T_i as initial values for the estimates $\hat{m}_i$ and (ii) apply IPF in the usual way (see Haberman 1974, pp. 64–73; Bishop *et al.*, 1975, pp. 83–97). The computer algorithm given by Haberman (1972) is one which can easily be used for this type of analysis. This algorithm is most commonly employed for fitting factorial models which are hierarchical; however, Haberman (1974, pp. 63–73) pointed out that this is equivalent to a cyclic ascent algorithm which may be used for fitting more general models and the general results of Darroch and Ratcliff (1972) hold. Another program FREQ, also given by Haberman (1979,

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Appendix), uses a Newton–Raphson approach for fitting a general log-linear model. This can also be used for fitting the exponential model described above.

3. Models for Survivorship

A great deal of work has recently been done on the analysis of censored survival data, much of it inspired by the studies of Cox (1972) (see Prentice and Kalbfleisch, 1979, for a review). The Cox model assumes that the hazard function has a log-linear form, i.e. $\lambda(t;\mathbf{X})=\exp\{\alpha^*(t)+\mathbf{X}'\boldsymbol{\beta}\}$. This hazard function can be approximated by dividing the period of follow-up into K intervals $(\tau_1, \tau_2]$, $(\tau_2, \tau_3]$, ..., (τ_K, ∞) and assuming the hazard is constant during each interval. Hence, the function $\alpha^*(t)$ is approximated by a step function $\alpha^*(t)=\alpha_k$ for $\tau_k<\tau\leqslant\tau_{k+1}$. Such a model has been described by Holford (1976) and it gives rise to exponential survival during each interval. The kernel of the log likelihood for this model is

$$L=\sum_k \alpha_k n_{+k}+\sum_{ik} n_{ik}\mathbf{X}_i'\boldsymbol{\beta}-\sum_{ik}\exp(\alpha_k+\mathbf{X}_i'\boldsymbol{\beta})T_{ik}, \tag{3.1}$$

where n_{ik} is the observed number of failures and T_{ik} is the total length of follow-up for the ith subgroup during the kth interval. The similarity between (3.1) and (2.2) is obvious, and the applicability of log-linear models follows immediately. In this case one uses the number of failures n_{ik} and the total follow-up time T_{ik} for subgroup i during interval k in the way described in the previous section and illustrated in the example. This model also holds if the variables change from interval to interval, i.e. we use the variables $\mathbf{X}_{ik}$. Although this model is not identical to Cox's, one can obtain his parameter estimators for $\boldsymbol{\beta}$ by an appropriate selection of intervals when ties are absent (see Breslow, 1972; Holford, 1976)

An advantage in using this procedure is that sufficient statistics can be obtained from a large data set with one pass through the data. These statistics summarize the survival experience for an interval. Related to this problem is the approach given by Prentice and Gloeckler (1978), and the work of Brown (1975), Thompson (1977) and Mantel and Hankey (1978). The latter three papers make use of the logit model for the survival probabilities which are well-known to be log-linear models. Hence, the applicability of log-linear models for contingency tables is straightforward for cases where the covariates are categorical. Mitchell and Turnbull (1979) described the application of log-linear models to the analysis of data from survival/sacrifice experiments, while Koch, Johnson and Tolly (1972), Gehan and Siddiqui (1973) and Johnson and Koch (1978) utilized a weighted least squares approach to the analysis of grouped survival data.

4. Example

To illustrate the approach we consider the data in Table 1 on survival from lung cancer in males who were diagnosed in Connecticut during 1973. Two factors are of interest: histology and stage of disease, each grouped into three categories. These data were reported by C. Boyle in an essay written in partial fulfillment of the requirements of the MPH degree at Yale University in 1976. The period of follow-up was divided into a series of two-month intervals. Part (a) of the table gives the number of deaths in each interval and (b) the total follow-up time in months. Time is given to the nearest month. The data in this table give sufficient statistics for the survival model given in §3. This summary is for 539 subjects; however, once such a table has been formed, this method of analysis can just as easily handle any number of subjects.

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Table 1

Number of deaths and total follow-up (in months) during bimonthly follow-up intervals by histology and stage for males diagnosed in Connecticut during 1973

(a) Number of deaths

	Histology	I			II			III		
Interval	Stage	1	2	3	1	2	3	1	2	3
0–		9	12	42	5	4	28	1	1	19
2–		*2*	*7*	*26*	*2*	*3*	*19*	*1*	*1*	*11*
4–		9	5	12	3	5	10	1	3	7
6–		10	10	10	2	4	5	1	1	6
8–		1	4	5	2	2	0	0	0	3
10–		3	3	4	2	1	3	1	0	3
12–		1	4	1	2	4	2	0	2	3
14–		3	3	3	1	2	0	0	1	1
16–		2	3	3	0	1	1	2	0	1
18–		2	1	1	0	0	0	0	0	0
20–		1	0	2	1	3	0	0	0	0
22–		1	1	0	0	0	0	0	0	0
24–		0	0	1	1	0	0	0	0	0
26–		0	0	1	0	0	0	0	0	0
28–		0	0	0	1	0	0	0	0	0

(b) Total follow-up time (in months)

	Histology	I			II			III		
Interval	Stage	1	2	3	1	2	3	1	2	3
0–		157	134	212	77	71	130	21	22	101
2–		139	110	136	68	63	72	17	18	63
4–		126	96	90	63	58	42	14	14	43
6–		102	86	64	55	42	21	12	10	32
8–		88	66	47	50	35	14	10	8	21
10–		82	59	39	45	32	13	8	8	14
12–		76	51	29	42	28	7	6	6	10
14–		71	38	28	34	19	4	6	4	3
16–		62	21	19	30	14	4	5	2	2
18–		49	16	13	28	11	2	2	2	0
20–		42	12	9	25	9	2	2	2	0
22–		34	10	6	19	4	2	2	2	0
24–		19	4	4	15	2	1	0	0	0
26–		5	2	1	11	2	0	0	0	0
28–		4	4	0	8	4	0	0	0	0

Initially, the usual log-linear models were fitted by IPF using Table 1(b) as starting values for Table 1(a). Particular models are represented using the Bishop *et al.* (1975, p. 61) notation, so that $C_{12}C_3$ refers to a model which has an interaction between Variables 1 and 2, all main effects and the grand mean. The results of this analysis are summarized in Table 2. One can conclude from this that stage is certainly an important prognostic variable but histology does not contribute significant additional information on survival in these data.

The models considered in Table 2 do not include interactions with intervals but assume

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Table 2
Summary chi-square analysis for data in Table 1

Model*	df	G^2	Effect	Δdf	ΔG^2
$C_{12}C_3$	97	79.50			
$C_1C_2C_3$	101	82.39	Stage–histology interaction	4	2.89
C_1C_3	103	84.28	Histology adjusted for stage	2	1.89
C_2C_3	103	194.46	Stage adjusted for histology	2	112.07‡
C_3	105	203.31	Histology	2	8.85†
			Stage	2	119.03†

*The variables considered are: stage (1), histology (2) and interval (3).
†$P<0.05$
‡$P\ll 0.001$

instead a proportional hazards model. It would be easy to fit models including these interactions, but the number of deaths in some intervals is so small that interpretation of the results would be rather difficult. Instead, interaction of a linear term for period and the main effects for histology and stage were considered using Haberman's (1979, Appendix) program FREQ. The results gave chi square tests $\Delta G^2=2.37$ and $\Delta G^2=4.15$ for interaction of period with histology and stage, respectively, both with 2 df and both not significant.

5. Discussion

Models for events occurring in time, including the Poisson process, exponential and survival models, can be analyzed using log-linear models. This use of log-linear models is reminiscent of the situation where one is considering quasi-independence models (Bishop *et al.*, 1975, pp. 178–182). Quasi-independence can be represented by setting the values of T_i to be 0 or 1. However, for a general log-linear model we can have any known positive values of T_i. Of particular interest are the results of Darroch and Ratcliff (1972) showing the applicability of IPF to this situation. We have considered some specific applications in epidemiology and medicine where the more general results may be used.

The Poisson model used for analyzing rates has been fitted using an iterative indirect standardization (IIS) technique (see Mantel and Stark, 1968; Breslow and Day, 1975; Osborn, 1975; Gail, 1978). This algorithm is similar to IPF but not equivalent to it. Applying IPF to the examples given by these authors seems to give rates of convergence which are essentially the same as those obtained in IIS. An advantage of IPF is the availability of computer programs and the ease of representing higher-order models.

The application of log-linear models to the analysis of survival data is particularly well-suited for large data sets. When the covariables are categorical, sufficient statistics can be obtained in the form of a multidimensional contingency table with a single pass through the data. This method would be better than the usual methods in cases where the number of cells in the tables is considerably less than the number of observations. However, one must also consider the particular type of algorithm used for computation. Haberman (1974, p. 73) suggests that Newton–Raphson is more efficient than IPF when the number of parameters is not too large (less than about 28), while the converse is true for large numbers of parameters. For models with many parameters, such as the case where many high-order interactions are considered, IPF may be less expensive computationally. In this case one may use the tabular summary instead of the data because of the ease of using an IPF program which may be more efficient for the problem at hand if the

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only alternative is, say, Newton–Raphson. The trade-off here might be an increase in memory for a savings in CPU time.

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Since the original submission of this manuscript I have become aware of related unpublished work by others, in particular some work by Nan Laird and Donald Olivier on the use of log-linear models for the analysis of censored survival data. A referee also brought to my attention a manuscript by Aitkin and Clayton (1980), which makes use of GLIM (Nelder, 1977) to analyze censored survival data.

Résumé

On examine des modèles dans lesquels la variable analysée: le taux d'occurence d'un certain événement, est en relation log-linéaire avec les covariables. On montre alors que l'estimation des paramètres met en jeu la solution de systèmes d'équations identiques pour un processus de Poisson, une distribution exponentielle, un modèle de taux de survie ou un modèle log-linéaire généralisé. Ceci nous rend donc capable d'utiliser les algorithmes d'ajustement log-linéaire (par exemple l'ajustement proportionel itératif: A.P.I.) pour l'analyse de proportions ou de taux de survie.

References

Aitkin, M. and Clayton, D. (1980). The fitting of exponential, Weibull and extreme value distributions to complex censored survival data using GLIM. *Applied Statistics*. Accepted for publication.

Armitage, P. (1966). The chi-square test for heterogeneity of proportions after adjustment for stratification. *Journal of the Royal Statistical Society, Series B* **28,** 150–163.

Bishop, Y. M. M., Fienberg, S. F. and Holland, P. W. (1975). *Discrete Multivariate Analysis*. Cambridge, Massachusetts: M.I.T. Press.

Bjarnason, O., Day, N., Snaedal, J. and Tulinius, H. (1974). The effect of year of birth on the breast cancer age–incidence curve in Ireland. *International Journal of Cancer* **13,** 689–696.

Breslow, N. (1972). Covariance analysis of censored survival data. *Biometrics* **30,** 89–99.

Breslow, N. E. and Day, N. E. (1975). Indirect standardization and multivariate models for rates, with reference to the age adjustment of cancer incidence and relative frequency data. *Journal of Chronic Diseases* **28,** 289–303.

Brown, C. C. (1975). On the use of indicator variables for studying the time-dependence of parameters in a response–time model. *Biometrics* **31,** 863–872.

Cox, D. R. (1972). Regression models and life-tables (with discussion). *Journal of the Royal Statistical Society, Series B* **34,** 187–220.

Darroch, J. N. and Ratcliff, D. (1972). Generalized iterative scaling for log-linear models. *Annals of Mathematical Statistics* **43,** 1470–1480.

Feigl, P. and Zelen, M. (1965). Estimation of exponential survival probabilities with concomitant information. *Biometrics* **21,** 826–838.

Gail, M. (1978). The analysis of heterogeneity for indirect standardized mortality ratios. *Journal of the Royal Statistical Society, Series A* **141,** 224–234.

Gehan, E. A. and Siddiqui, M. M. (1973). Simple regression methods for survival time studies. *Journal of the American Statistical Association* **68,** 848–856.

Glasser, M. (1967). Exponential survival with covariance. *Journal of the American Statistical Association* **62,** 561–568.

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Haberman, S. J. (1972). Log-linear fit for contingency tables (Algorithm AS51). *Applied Statistics* **21,** 218–225.

Haberman, S. J. (1974). *The Analysis of Frequency Data.* Chicago: University of Chicago Press.

Haberman, S. J. (1978). *Analysis of Qualitative Data. Vol.* I, *Introductory Topics.* New York: Academic Press.

Haberman, S. J. (1979). *Analysis of Qualitative Data. Vol.* II, *New Developments.* New York: Academic Press.

Holford, T. R. (1976). Life tables with concomitant information. *Biometrics* **32,** 587–597.

Kilpatrick, S. J. (1962). Occupational mortality indices. *Population Studies* **16,** 175–187.

Koch, G. G., Johnson, W. C. and Tolley H. D. (1972). A linear models approach to the analysis of survival and extent of disease in multidimensional contingency tables. *Journal of the American Statistical Association* **67,** 783–796.

Johnson, W. D. and Koch, G. G. (1978). Linear models analysis of competing risks for grouped survival times. *International Statistical Review* **46,** 21–51.

Mantel, N. and Hankey, B. F. (1978). A logistic regression analysis of response time data where the hazard function is time dependent. *Communications in Statistics—Theory and Methods* **A7** (4), 333–347.

Mantel, N. and Stark, C. R. (1968). Computation of indirect adjusted rates in the presence of confounding. *Biometrics* **24,** 997–1005.

Mitchell, T. J. and Turnbull, B. W. (1979). Log-linear models in the analysis of disease prevalence data from survival/sacrifice experiments. *Biometrics* **35,** 221–234.

Nelder, J. A. (1977). *General Linear Interactive Modelling* (*GLIM*). *Release* 3. Oxford: Numerical Algorithms Group.

Osborn, J. (1975). A multiplicative model for the analysis of vital statistical rates. *Applied Statistics* **24,** 75–84.

Prentice, R. L. and Gloeckler, L. A. (1978). Regression analysis of grouped survival data with application to breast cancer data. *Biometrics* **34,** 57–67.

Prentice, R. L. and Kalbfleisch, J. D. (1979). Hazard rate models with covariates. *Biometrics* **35,** 25–39.

Thompson, W. A. (1977). On the treatment of grouped observations in life studies. *Biometrics* **33,** 463–470.

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